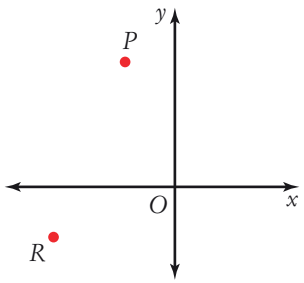
WORKSHEET

Chapter 4: Addition and subtraction in the plane

1 The points *P*, *Q* and *R* correspond to the complex numbers *p*, *q* and *r*. *PQRO* forms a parallelogram.



Use the diagram shown for the following:

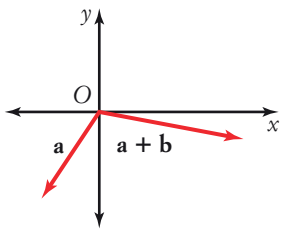
a Plot the point *Q*.

b Express the vector **q** in terms of **p** and **r**.

c Express the vector $\overrightarrow{RP}$ in terms of **p** and **r**.

d Plot the point *S* that corresponds to the vector **p** − **r**.

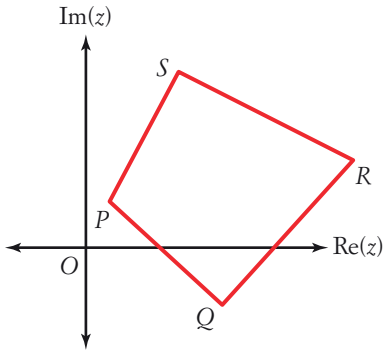
2 The vectors **a** and **a** + **b** corresponding to the complex numbers *a* and *a* + *b* are plotted on an Argand diagram.



Use the diagram to draw the vector **b** and plot the location of the point *B* representing *b*.

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3 Consider the vertices of the quadrilateral $PQRS$ representing the complex numbers z_1, z_2, z_3 and z_4 respectively.



a Express each vector in terms of any two of z_1, z_2, z_3 or z_4:

i $\overrightarrow{PS}$

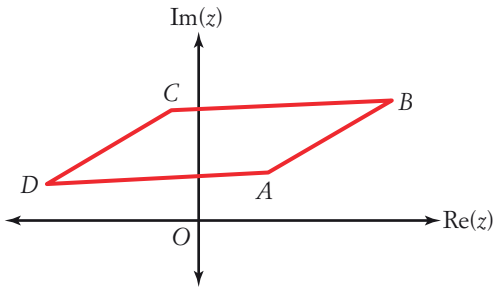
ii $\overrightarrow{QR}$

iii $\overrightarrow{PR}$

iv $\overrightarrow{SQ}$

b The midpoint M of SQ is represented by the complex number m. Find m in terms of z_1, z_2, z_3 or z_4.

4 In the diagram, $ABCD$ is a parallelogram. The points A, B, C and D represent the complex numbers z, w, u and v respectively.



a Find in terms of z, w, u or v:

i $\overrightarrow{AD}$

ii $\overrightarrow{BC}$

iii $\overrightarrow{CA}$

iv $\overrightarrow{DB}$

b Find the complex number m represented by the midpoint M of AC in terms of z, w, u or v.

5 A, B, C and D representing the complex numbers z_1, z_2, z_3 and z_4 respectively form a non-equal sided quadrilateral in the complex plane. State what type of quadrilateral $ABCD$ is, given

$|z_2 - z_1| = |z_4 - z_1|$

and $|z_3 - z_2| = |z_4 - z_3|$.

6 P, Q, R and S representing the complex numbers w_1, w_2, w_3 and w_4 respectively form a quadrilateral in the complex plane. State what type of quadrilateral $PQRS$ is, given:

$w_2 - w_1 = w_3 - w_4$

$w_4 - w_1 = w_3 - w_2$

and $|w_4 - w_1| = |w_2 - w_1|$

7 Consider the points W and Z representing the complex numbers $w = -2 + 7i$ and $z = 6 + 2i$ on the Argand plane. O is the origin and V is another point on the plane such that $OZVW$ is a parallelogram.

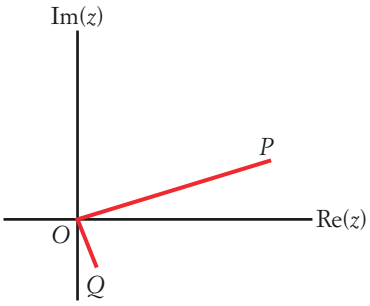
a Find the complex number v that represents the point V.

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b Find the length of OV.

c Find the length of ZW.

8 Consider the complex numbers $p = 5\sqrt{3} + 5i$ and $q = 1 - i\sqrt{3}$ represented by the points P and Q on an Argand diagram such that $OPQR$ is a quadrilateral.



a Plot the point R on the diagram and such that $OPQR$ is a parallelogram.

b Find the complex number r represented by R.

c Calculate $|p|$.

d Calculate $|q|$.

e Calculate $|r|$.

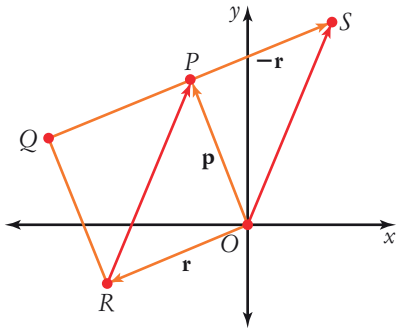
f Calculate $|p - q|$.

g What type of quadrilateral is $OPQR$?

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Answers

1



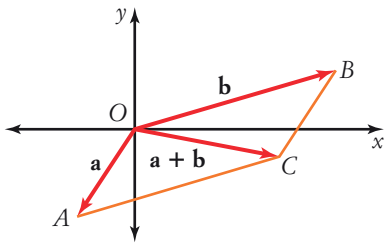
a See diagram

b $\mathbf{q} = \mathbf{p} + \mathbf{r}$

c $\overrightarrow{RP} = \mathbf{p} - \mathbf{r}$

d See diagram

2



3 a i $\overrightarrow{PS} = z_4 - z_1$

ii $\overrightarrow{QR} = z_3 - z_2$

iii $\overrightarrow{PR} = z_3 - z_1$

iv $\overrightarrow{SQ} = z_2 - z_4$

b $m = z_4 + \frac{1}{2}(z_4 - z_2)$ (many other answers)

4 a i $\overrightarrow{AD} = v - z$

ii $\overrightarrow{BC} = u - w$

iii $\overrightarrow{CA} = z - u$

iv $\overrightarrow{DB} = w - v$

b $m = \frac{1}{2}(v + w)$

5 Kite, adjacent sides are equal.

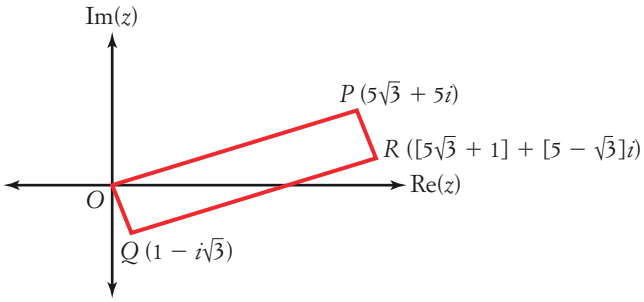
6 Rhombus, opposite sides are parallel and one pair of adjacent sides is equal.

7 a $z = 4 + 8i$

b $4\sqrt{5}$

c 10

8 a



b $r = [(5\sqrt{3} + 1) + (5 - i\sqrt{3})i]$

c $|p| = 10$

d $|q| = 2$

e $|r| = 2\sqrt{26}$

f $|p - q| = 2\sqrt{26}$

g $OPQR$ is a rectangle.

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